

Unit 4: Input Devices

Learning Objectives

- Understand the types of input devices.
- Learn their advantages and disadvantages.
- Explore practical applications in various domains.



- **Computer cannot understanding our language because it understands only machine language and it can be possible using some devices are called input devices**
- ***Input device* captures information and translates it into a form that can be processed and used by other parts of your computer**



Introduction to Input Devices

- Input devices are hardware components used to provide data and control signals to a computer. They are essential for human-computer interaction.



Types of Input Devices

Keyboards

Pointing devices

Game controllers

Scanners

Styluses

Microphones

Digital cameras

Web cams

LIST OF SOME INPUT DEVICES:

- **Mouse**
- **Keyboard**
- **Trackball**
- **Space ball**
- **Joystick**
- **Digitizer**
- **Dials**
- **Button boxes**
- **Data gloves**
- **Touch panels**
- **Image scanners**
- **Voice systems**



KEYBOARD



- Typewriter keys
- Function keys
- Enter keys
- System keys
- Numeric keypad
- Other
- Application key
- Cursor control keys

KEYBOARD

- **Keyboard is the most commonly used input devices.**
- **They allow data entry into a computer system by pressing a set of keys neatly mounted on a keyboard**
- **Contains some extra command keys and function keys that affect the operation of computer or keyboard itself**
- **Types of keyboard :**
 - (1). Standard keyboard
 - (2). Multimedia keyboard
 - (3). Wireless keyboard
- **In Graphics:**
- **Used to provide screen coordinates**
- **Menu selection**
- **Gaming controls**
- **And FOR entering many graphics function**

Keyboards

- Description: Standard, ergonomic, and virtual keyboards.
- Advantages: Familiar interface, easy to use.
- Disadvantages: Limited to text entry, physical strain.
- Applications: Text entry in software applications.

KEYBOARD TYPES:

Foldable type



Wireless



Standard



Latest



Multimedia





Categories of Input Devices

- • Manual Input Devices: Keyboards, Mice
- • Automatic Input Devices: OCR, Magnetic Readers
- • Specialized Input Devices: Light Pens, Touch Screens



MOUSE

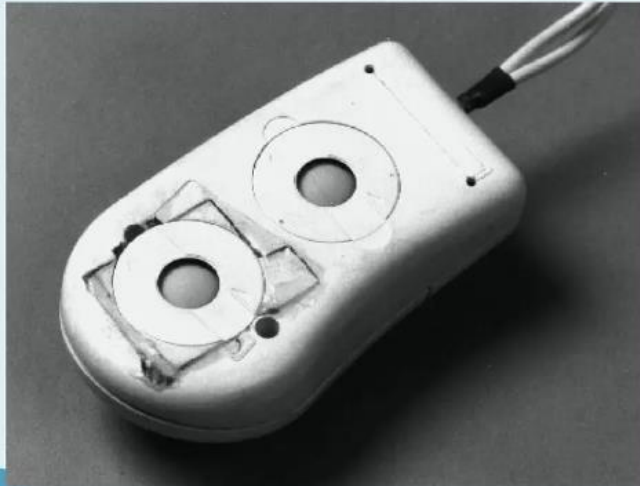
- **Hand-held BOX** used to position the screen cursor
- **Mouse** is the most popular point-and-draw device
- **Wheels or Rollers** (*now-a-days Laser lights*) on the bottom are used to record the position of the screen
- Generally there are two or three buttons, used for operations like recording of the cursor positions or invoking of a function
- In order to increase the number of **INPUT** parameters, additional devices can be included

Types of mouse:

- (1). Serial mouse
- (2).PS/2 mouse
- (3).Optical mouse
- (4).Wireless mouse
- (5).USB mouse



MOUSE



Mice

- Description: Optical and mechanical types.
- Advantages: Intuitive point-and-click operation.
- Disadvantages: Requires flat surface, limited precision in design work.
- Applications: GUI navigation, gaming, design work.

TYPES OF MOUSE



TRACKBALL



- It is a 2D positioning device
- It consists a ball held by a socket containing sensors to detect the rotation of ball about TWO axis
- User rolls the ball to move the cursor
- They are often mounted on devices such as keyboards, Z-mouse ETC



SPACEBALLS

- It provides SIX degrees of freedom
- It is a fix device
- Movement detection is done using strain gauges
- Cursor can move in any direction
- It is more efficient then trackball



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- **Used in CAD workstations**
 - **In animation**
 - **Sometimes on special Workstations such as the radar consoles in air-traffic control room**
 - **In Gaming consoles**
 - **People with a mobility impairment use trackballs as an assistive technology input device**

JOYSTICK



- Consists of a stick pivoted on a base
- Used to steer the screen cursor
- It also has one or two PUSH buttons as input switches to perform certain actions
- Most joystick are 2D, but 3D do exist
- Distance moved from the CENTER position corresponds to the screen cursor movement in that direction
- In Gaming consoles
- In 3D animation
- Used to drive machines like cranes, mining trucks, hydraulics ETC
- Used as assistive technology pointing device such as in Electronic wheelchairs





DATA GLOVE

- Used to grasp a “virtual” object
- Uses sensors to detect the Hand and finger motion
- Electromagnetic coupling between signals provides information about the position and orientation of the HAND

Applications:

- In 3D animation movies
- Visual effects
- Gestures can be categorized into useful information, such as to recognize Sign Language or other symbolic functions
- 3D Virtual environment Games



DIGITIZERS



- Common device for drawing, painting, or interactively selecting coordinate positions on an object



- Typically, it is used to scan an Object and to input discrete coordinate positions

ONE TYPE of Digitizer is the Graphics Tablet



GRAPHICS TABLET



- A **graphics tablet** is a computer input device that allows one to hand-draw images and graphics, similar to the way one draws images with a pencil and paper. These tablets may also be used to capture data or handwritten signatures

- The common drawing TOOLS used to draw are **HAND CURSOR** and **STYLUS**

- A **STYLUS** is a pencil-shaped pointing device

This Digitizing system uses electromagnetic resonance to detect the 3D positions of the STYLUS

This allows one to produce different shades of brush strokes with different pressure on its surface

Acoustic Tablets use sound waves to detect the position of the STYLUS

3D Digitizers use sonic or electromagnetic transmissions to record positions.

- It uses the same mechanism used in the Data Gloves.

APPLICATIONS:

- Used in generating Computer generated graphic images
- Used in creating characters for Animation
- In Technical drawings and CAD
- Used for Handwriting recognition





IMAGE SCANNERS

- In computing, a **scanner** is a device that optically scans images, printed text, handwriting, or an object, and converts it to a digital image
- When the scanning is performed, the gradation of gray scale or colors are recorded and stored in an array
- Once scanned, any kind of transformations can be applied to the object image



LIGHT PENS



- A **light pen** is a computer input device in the form of a light-sensitive wand used in conjunction with a computer's CRT TV set or monitor
- Allows the users to point to displayed objects and to draw objects on screen
- The position points are highly accurate and sensitive
- It generates electrical pulse which records the position of the electron beam
- Not very much popular



TOUCH PANEL

- Touch panels allow displayed objects or screen positions to be selected with the touch of a finger.
- One appln of touch panels is for the selection of processing options that are represented with graphical icons.
- Touch input can be recorded using optical, electrical or acoustical methods.
- Optical touch panels employ a line of infrared LEDs along one vertical edge and along one horizontal edge of the frame. The opposite vertical and horizontal edges contain light detectors.
- These detectors are used to record which 27 Light Pens 29 beams are interrupted when the panel is touched. The 2 crossing beams that are interrupted identify the horizontal and vertical coordinates of the screen position selected.



VOICE SYSTEM

- Speech recognizers are used in some graphics work stations as i/p devices to accept voice commands. The voice system i/p can be used to initiate graphics operations.
- These systems operate by matching an i/p against a predefined dictionary of words and phrases.
- Later when a voice command is given, the system searches the dictionary for a match. Voice i/p is typically spoken into a microphone mounted on a handset.



Light Pens

- Description: Device for drawing or selecting on a screen.
- Advantages: Direct interaction with the screen.
- Disadvantages: Requires special display, not ergonomic.
- Applications: CAD design, graphical editing.

Optical Character Recognition (OCR)

- Description: Converts printed or handwritten text into digital form.
- Advantages: Saves time, reduces manual effort.
- Disadvantages: Accuracy depends on text quality.
- Applications: Digitizing documents, automating data entry.

Mark Readers

- Description: Reads data from marked forms (e.g., MCQ sheets).
- Advantages: High-speed, cost-effective for large-scale data.
- Disadvantages: Limited to specific forms.
- Applications: Exam scoring, surveys.

Magnetic Stripe Readers

- Description: Reads data from magnetic strips (e.g., credit cards).
- Advantages: Fast data input, reliable.
- Disadvantages: Vulnerable to physical damage, security concerns.
- Applications: Banking, retail, access control.

Summary

- • Overview of common input devices.
- • Their advantages, disadvantages, and applications.
- • Importance in various fields.